

ASMITHA U

AI ENGINEER | SOFTWARE ENGINEER | LLMs, RAG & AGENTIC SYSTEMS

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PROFESSIONAL SUMMARY

Software engineer with **3+ years** at Palo Alto Networks, progressing from Software Engineer Intern to **Staff Software Engineer**, building distributed backend services and cloud infrastructure across **GCP, AWS, and Kubernetes**. Pursuing an MSc in Enterprise Artificial Intelligence at NTU, with hands-on work in **LLM applications, RAG, agentic workflows**, multimodal document understanding, and evaluation of accuracy, latency, cost, and failure modes.

PROFESSIONAL EXPERIENCE

Staff Software Engineer - Palo Alto Networks

Jun 2025 - Jun 2026

- Engineered an event-driven device lifecycle workflow using GCP Pub/Sub across enterprise systems, preventing duplicate processing across concurrent REST and message-driven paths.
- Resolved customer-impacting production incidents involving lifecycle state drift, software-version conflicts, and distributed backend failures.

Software Engineer - Palo Alto Networks

Aug 2023 - May 2025

- Designed a multi-cloud storage abstraction for Amazon S3 and Google Cloud Storage, reducing estimated storage and egress costs by **20-50%** while reducing vendor lock-in.
- Migrated legacy microservices from virtual machines to Kubernetes and improved NGINX, secrets management, and CI/CD deployment practices.
- Developed APIs and batch workflows for enterprise software lifecycle management, automating end-of-life deallocation and version tracking.
- Increased unit-test coverage by **30-40%** using JUnit and WireMock, strengthening API validation and system reliability.

Software Engineer Intern - Palo Alto Networks

Jan 2023 - Jul 2023

- Built a Python framework to migrate BigQuery data across GCP regions, improving transfer efficiency by **30%** with **zero data loss**.
- Automated JSON-to-Excel security-scan reporting in Python, reducing manual effort by **60%**, and remediated Java vulnerabilities to improve compliance.
- Implemented MongoDB multi-region replica-set synchronisation using AWS volume snapshots, reducing downtime by **40%** and maintaining data consistency across regions.

EDUCATION

Nanyang Technological University, Singapore - MSc in Enterprise Artificial Intelligence

Aug 2026 - Present

Coursework: Generative AI, Agentic AI, Enterprise AI Solutions, AI Project Management

Amrita Vishwa Vidyapeetham | B.Tech, Computer Science Engineering (AI)

Sep 2019 - May 2023

CGPA: 9.22/10 | Coursework: Machine Learning, Deep Learning, Computer Vision, NLP

AI PROJECTS

AI Corporate Expense Compliance Checker

Designed and prototyped a multimodal receipt-review pipeline combining OCR, lightweight RAG over company policy, and LLM reasoning with gated actions and human escalation; evaluated **accuracy, latency, cost, and failure cases**.

Adaptive AI for Enterprise NDA Compliance Review

Designed an adaptive RAG system on ContractNLI for clause-level retrieval and three-way classification, with low-confidence human review; defined **Macro-F1, risk recall, Retrieval Recall@K, latency, and cost** measures.

Anomaly Detection for System Logs

Applied ANN, CNN, RNN, Isolation Forest and Decision Trees to operational log data for real-time anomaly detection and system monitoring.

PUBLICATIONS & AWARDS

- Analysis of Stop Consonants and Vowels in Indian Languages: A Multifractal Approach - IEEE ICMSS 2021 (Best Paper Award)
- DistilRoBERTa-Based Sentence Embedding for Rhetorical Role Labelling of Legal Case Documents - FIRE 2021
- Coconut Tree Detection Using Deep Learning Models - Springer, 2023
- Ensemble Deep Learning Models for Vehicle Classification in Motorized Traffic Analysis - Springer, 2023
- Collaboration Team Award, Palo Alto Networks - recognized for driving cross-functional alignment across Engineering, QA, and Product teams; mentored junior engineers.

CERTIFICATIONS

Google Cloud Certified - Professional Cloud Architect | Jan 2026 - Jan 2028

TECHNICAL SKILLS

AI/ML: Generative AI, LLMs, RAG, Agentic AI, Machine Learning, Deep Learning, NLP

Programming: Python, Java, SQL, Bash

Cloud & Backend: AWS, GCP, Kubernetes, Docker, Microservices, REST APIs, MongoDB, BigQuery

Engineering: Git, Jenkins, CI/CD, JUnit, WireMock, Linux, Kibana, Datadog, Grafana

Product & Collaboration: Stakeholder Management, Cross-functional Collaboration, Technical Prioritization, Agile